



Intellectual capital and business start-up success

Iñaki Peña

ESTE, Deustuko, Unibertsitatea, Donostia, Spain

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Abstract *While start-up firms create a substantial economic impact on most economies, the failure rate of start-up firms seems to remain high over time. Few authors have examined the influence of intellectual capital management on business performance, and when it has been examined, the focus has been on large, mature and public companies. The purpose of this study is to analyze the extent to which IC assets are associated with new firm survival and growth. Results from this study suggest that the human capital of the entrepreneur (i.e. education, business experience and level of motivation), organizational capital (i.e. firm capacity to adapt quickly to changes and the ability to implement successful strategies), and relational capital (i.e. development of productive business networks and an immediate access to critical stakeholders) are important intangible assets, which seem to be related positively to venture performance.*

Introduction

There is a wide feeling of agreement about the significant impact that new and small firms create on the whole economy. Reynolds (1987) found that 30 percent of new firms provided between 60-80 percent of the new jobs, sales and exports in the USA. Policy makers believe that new businesses are not only a driving force of local economies, but also constitute a motor of change in several industry sectors. With all these facts in mind, it is not surprising to observe the emergence of recent public policies which aim to encourage the creation of new firms (i.e. business incubators, scientific parks, etc.). Numerous studies suggest, on the other hand, that young firms are fragile and the failure rate of business start-ups remains high over time. The study by Van de Ven *et al.* (1984) concludes that about 54 percent of new businesses survive a year and a half, and only 25 percent survive six years. Certainly, failure rates are not homogeneous across industries and the service sector shows the higher percentage of failed ventures, followed by the retail, manufacturing and high-tech sectors respectively. Cooper and Dunkelberg (1981) found that 71 percent of service and retail firms failed after four to five years. After considering all these facts the study of determinants of the success of new firms really becomes an opportune topic.

Throughout the last decade, we have witnessed frequent and considerable discrepancies between book and market values of companies around the world. An increasing number of scholars and business practitioners believe that this gap between book and market values can be partially explained by the value accrued to intellectual capital and other intangible assets. While previous



research on intellectual capital has focused mainly on topics related to the definition, classification and measurement of intellectual capital (IC), few authors have examined the relationship between IC assets and business performance.

The recent work by Bontis *et al.* (2000) suggests the existence of a significant positive relationship of intellectual capital with business performance, regardless of the industry sector. The authors found that organizations' efforts to manage knowledge and develop their structural capital led to relatively higher business performance. To support this line of thinking, Sánchez *et al.* (2000) state that intangible assets are important in management processes since they are considered to be a critical source of sustainable competitive advantage. Roos *et al.* (1997) examine the linkage between IC and the value creation process of a company from a dynamic perspective. The authors provide important insights regarding the influence of IC tools on the value of an organization.

Despite the relevance of these recent works, still more empirical work is needed to confirm the link between IC assets and sustainable supra-normal profits (Bontis *et al.*, 2000). Apart from shareholder value creation, an IC management approach can be extremely useful for other types of companies such as non-public companies, family businesses, and new ventures. The focus of this study is precisely the analysis of the relationship between the intellectual capital of a start-up firm and new venture success. The adoption of an IC managerial perspective on business start-ups is expected to help entrepreneurs overcome the difficult initial years of their organizations. Business start-up success can be partially explained by human, organizational and relational capital elements. Each of these factors deals with intangible elements such as skills of entrepreneurs, strategic decision making, market adaptation, networking ability with external stakeholders, and so on. We assume that, like for stock-market companies, IC assets play also a key role in explaining the survival and growth of new firms.

Considering both the economic relevance and fragility of business start-ups, the purpose of this study is to examine the extent to which IC assets explain new firm survival and growth. This work is an attempt to sort out IC assets into three conventional blocks (i.e. human capital, organizational capital and relational capital) and to assess the relationship of each block with new venture performance by examining data from a sample of business start-ups. Findings from this study are expected to shed some light on IC determinants of venture success.

The article consists of five additional sections. The following section is a brief review of relevant literature on distinct IC frameworks and determinants of new firm survival and growth. A simplified conceptual model, which relates IC elements to start-up success, is proposed in part three. Section four deals with the data and methodology used in our empirical work. After the discussion of results, the study concludes with a summary of main findings and implications.

IC frameworks and firm success factors

We will present a brief description of relevant studies undertaken in two research streams, intellectual capital and entrepreneurship, with the aim of developing a theoretical basis for the conceptual model of the next section. We will start with the exposition of recent IC works and will follow with a short outline about the determinants of new firm survival and growth found in the literature.

Recent IC frameworks

Recent IC frameworks have emphasized three characteristics. First, the increasing importance of IC management requires a shift in the way corporate managers run their organizations. Knowledge has become a critical resource that firms must master in today's competitive landscape, and in order to manage knowledge, corporate managers need information that is not supplied by traditional accounting systems and financial reports. Therefore, non-financial indicators to measure intangible assets are eagerly needed[1].

Second, an IC framework must respect and comprehend each firm's idiosyncrasy. The framework must be tailored to meet the specific information needs of a particular organization in order to improve its own strategic decision making. And third, the framework has to be understood from a dynamic perspective. We must recognize the existence of a temporal dimension behind all the complex relationships among different intangible assets. The interaction among different IC elements experiences a permanent change as firms evolve over time. Furthermore, rather than a unidirectional causality, we will probably expect a bi-directional causality relationship between financial and non-financial indicators, human-structural-relational IC elements, IC asset-performance effects, etc.

Brennan and Connell (2000) provide a thorough review of relevant IC frameworks. They distinguish basically two types of frameworks: frameworks to classify intellectual capital assets and frameworks to manage intellectual capital. Our focus will be the former and we will center our attention on those studies which have proposed IC-related conceptual models during recent years.

Roos *et al.* (1997) developed a comprehensive framework which contains two main blocks: the "thinking IC" and the "non-thinking IC". The first block represents human capital and the second block describes the structural capital. The authors state that while the value of human capital stems from the competence, attitude and intellectual agility of individuals from an organization, the value of structural capital can be explained by the relationships of a firm with external actors. Specifically, structural capital value stems from the organizational value derived, on the one hand, from internal infrastructure, processes and culture, and on the other hand, from renewal and developmental strategies. Rather than taking a static view of each of these forms of capital, we must understand the existence of changes in the stocks of capital and consider the "flows" of capital or interactions among all the forms of capital described earlier.

A quite similar IC framework was developed by Edvinsson and Malone (1997). They point out that “whereas traditional accounting is, by design a presentational tool of the past, IC is, by default, primarily a navigational tool of the future”. The authors cite an example of an IC navigation tool known as the Skandia Navigator, which is an IC framework with five areas of focus: financial, customer, process, renewal and development and human focus. The financial focus represents the past of the firm, whereas the process and customer focus describe the present activities of a company. The future is embedded in a renewal and development focus, which in addition to preparing the path that a firm must follow, also addresses the issue of abandoning the obsolete part of an organization. We find the human focus at the center of this framework, because people are the “soul” of the organization. This model also establishes linkages among the five areas of focus like the flows among different forms of capital proposed by Roos *et al.* in their framework.

It is not easy to state accurately in few words what we understand by IC. Stewart (1997) comes up with a definition of IC as:

... the sum of everything everybody in a company knows that gives it a competitive edge ...
IC is intellectual material knowledge, experience, intellectual property, information, ... , that can be put to use to create wealth.

He distinguishes two IC forms: the semi-permanent body of knowledge and the tools that augment the body of knowledge. Stewart (1997) and Sveiby (1997) coincide with the frameworks described earlier in that there exist human, internal and external IC components. Even though each author names each type of capital in a different manner, conceptually their meanings are the same.

The model proposed by Sánchez *et al.* (2000) differentiates between intangible resources (i.e. static dimension) and intangible activities (i.e. dynamic dimension). They also establish in their framework the conventional distinction between human, structural and relational capital and highlight the existing interaction among these three types of IC. An information system for the management of intangibles should comprise three phases: the identification of critical intangible resources to the strategic objectives of an organization, the development of useful indicators for intangible resources and activities, and finally, the monitoring of progress of these intangibles.

In sum, one can observe from all these IC frameworks the distinction between different forms of IC, namely IC stocks (i.e. such as human, structural and relational capital), and the existence of flows among them. In the majority of IC frameworks, the human capital component is complemented with intangible assets dealing with the internal structure and organization of a firm, and with the external relationship of an organization with its stakeholders.

Determinants of new firm survival and growth

It is well known that new firms face serious difficulties to survive their initial years of operations. Lack of business experience, strong industry competition, and small firm fragility are some of the most often cited arguments that justify

business failure. The intent of this section is to organize those determinants of firm survival and growth that are found in the literature. We must note that these determinants include both tangible and intangible factors. From all these factors, we have come up with the following classification: the human capital of the entrepreneur, firm start-up characteristics, industry environment, geographical location, and public policies. We appreciate that these factors correspond approximately to the fundamental elements contained in all the IC frameworks introduced earlier, that is, the human capital, firm internal capital (i.e. organizational or structural capital), and external capital (i.e. relational or customer capital).

Human capital can be defined as the accumulation of personal attributes (i.e. knowledge, abilities, personality, health etc.) that allow human beings to function. More investment in human capital is expected to yield a higher productivity of individuals. While the economic value of human capital can not be questioned, an important concern among entrepreneurs is what type and how much human capital is required to create a successful company. The talent, experience and motivation that entrepreneurs bring to an organization can be expected to determine the success level achieved by firms. Numerous studies indicate that the entrepreneur's level of education is positively associated with firm survival and growth (Bates, 1995; Hay and Ross, 1989; Cooper *et al.*, 1989; Van de Ven *et al.*, 1984), but experience seems to be another no less important human capital element. The knowledge base acquired in schools, college or continuous education programs can be enriched with business experiences. Some of these experiences are associated with the prior involvement of entrepreneurs in managerial positions. Studies show that firms are more likely to succeed when entrepreneurs have occupied a management position in a former company (Stuart and Abetti, 1990; Cooper *et al.*, 1989; Doutriaux and Simyar, 1987). Another useful experience seems to be the prior creation of companies. In this regard, some authors have centered their attention on prior start-up experience or the experience gained in owning a firm in the past (Dyke *et al.*, 1992; Doutriaux and Simyar, 1987; Stuart and Abetti, 1990). The findings are quite consistent and reveal that entrepreneurial start-up experience is expected to increase the odds of venture success.

Other not so profound experiences seem to be helpful for building successful companies. For instance, entrepreneurs whose parents, relatives or friends have gone through a business start-up process are acquainted with the endeavors of building a firm. By sharing the concerns of people close to them, entrepreneurs learn the intricacies of creating a business. Sometimes, entrepreneurs start a new firm based on the industry knowledge acquired from a previous job (i.e. knowledge about customers, suppliers, industry regulation, subsidies etc.). Empirical evidence supports that all these less intense experiences also matter to explain business success (Duchesneau and Gartner, 1990; Dunkelberg *et al.*, 1987; Cooper *et al.*, 1987).

In addition to knowledge, skills, abilities and experiences we must not ignore other intangibles such as the motivation with which entrepreneurs create their

companies. Personality attributes, attitude and ambition to achieve survival and growth goals for the new venture are also important. The motivation level of an entrepreneur who abandons a job, invests his or her own savings and devotes 55 hours a week in order to create a successful firm is very different from the motivation level of an unemployed entrepreneur who creates a company with government funds and devotes 25 hours a week to it. Therefore, entrepreneurs who feel strongly motivated to overcome the adversities of the initial years of business operations and to achieve challenging growth goals for their ventures are expected to show a better business performance.

In sum, the level of education, business experience and motivation of entrepreneurs constitute important human capital elements at the moment of starting up a new venture. Based on the results found in the literature, these IC assets embodied in entrepreneurs are expected to affect positively firm survival and growth.

In addition to the human component of an organization, venture success seems to be determined by the own characteristics of a firm. Typically, prior studies have examined the effects of firm age and size on business survival and growth. While older companies seem to be more likely to survive (Birley and Westhead, 1990; Kalleberg and Leicht, 1991), the relationship between firm size and performance is more ambiguous. Gibrat's Law states that the expected value of the increment to a firm's size in each period is proportional to the current size of the firm. However, empirical results do not consistently confirm that assertion and recent findings suggest that such a correlation can not be generalized (Agarwal and Audrestsch, 2001). As Geroski (1995) pointed out, growth rates of well-established corporations are random and, roughly speaking, do not seem to vary in any systematic way with firm size. Another significant firm characteristic is the legal form by which a company is created. Some studies reveal that business performance is considerably superior when the firm is started by a team of entrepreneurs, rather than by an individual (Woo *et al.*, 1989; Doutriaux and Simyar, 1987).

Business strategies are the result of managerial decisions, decisions that are constrained to the resources of a firm. A larger amount of financial capital provides an entrepreneur with more flexibility to undertake a wider array of strategies, and therefore, would influence positively business performance (Duchesneau and Gartner, 1990; Cooper *et al.*, 1989). The strategic decision making of companies is an organizational intangible that is supposed to affect directly the progress of a firm. Sandberg and Hofer (1987) found that firms which emphasized a "product differentiation" type of strategy outperformed the rest of start-up companies, whereas other studies showed that business strategies oriented towards the production of high quality products were related positively to firm survival and growth (Kalleberg and Leicht, 1991; O'Neil and Duker, 1986).

Besides internal factors, such as human capital and firm characteristics mentioned earlier, there are also external factors affecting the future wealth of start-up businesses. Some external factors have to do with location-specific

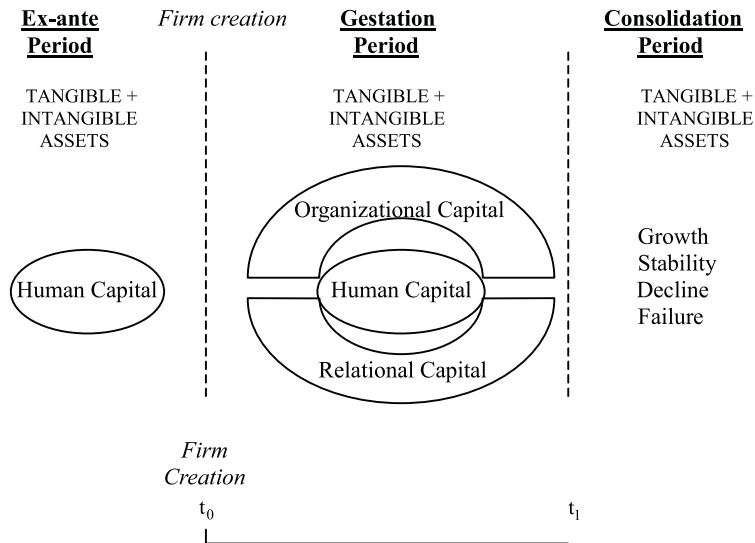
advantages or industry-specific advantages. We often hear success stories of high technology companies from the Silicon Valley area, telecommunications companies from Scandinavian countries, automotive companies from Japan, and so on. Certainly, the selection of a particular geographical location to build a business may be critical so long as advantages from network opportunities (i.e. with suppliers, customers, competitors, financial institutions, consultants etc.) and agglomeration economies (i.e. spill-over effects) translate into higher profits. Since all the markets do not exert the same competitive pressure, the choice of the industry sector also can be crucial. In the study by Sandberg and Hofer (1987), companies operating in industries in continuous disequilibrium, development or growth stage, with product heterogeneity and high entry barriers were more likely to experience superior business performance. In our opinion, both industry and geographical choices would influence the development of the relational capital of an organization.

Another important external factor which also seems to affect young firm success, is the set of public policies aimed to encourage new business creation. A business incubator is a popular policy aimed to promote venture creation. It provides entrepreneurs with the expertise, networks and tools they need to make their ventures successful. The US National Business Incubation Association defines business incubation as “a dynamic process of business enterprise development. Incubators nurture young firms, helping them to survive and grow during the startup period when they are most vulnerable. Incubators provide hands-on management assistance, access to financing and orchestrated exposure to critical business or technical support services. They also offer entrepreneurial firms shared office services, access to equipment, flexible leases and expandable space – all under one roof. An incubation program’s main goal is to produce successful graduates – businesses that are financially viable and freestanding when they leave the incubator, usually in two to three years”. In short, business incubators contain both tangible (i.e. space, equipment etc.) and intangible (i.e. formation programs, technical assistance, networking possibilities, entrepreneurial camaraderie) elements which help start-up firms overcome their difficult initial years (Mian, 1997).

After having analyzed in the literature different IC frameworks and the determinants of venture success, we propose a conceptual model, in which the relationship between IC components and firm start-up performance is examined. This conceptual model must be understood as an exploratory, rather than prescriptive, framework.

An IC framework for firm start-up

It is well known that, rather than entry barriers, firms often face barriers to survive during their initial years of operation. Our main hypothesis is that start-up firm performance will depend to a large extent on the effectiveness of IC management achieved by entrepreneurs during the gestation period. We will examine companies from a lifetime perspective, and correspondingly, we will distinguish three stages (see Figure 1).



Business start-up success

187

Figure 1.
IC framework for
business start-up

Usually, there exists an *ex ante* period in which the entrepreneur develops an idea and decides to create a company. In this stage, we only recognize the human capital IC component since the firm has not been created yet. The knowledge, experience and attitude embedded in an entrepreneur become critical not only to start a business, but also to ensure the endurance of the venture.

The second stage is the gestation period. Once the company is created, it gets immersed in a complex process, in which entrepreneurs must make a considerable effort to adjust the company to market conditions, and most importantly, to build a successful organization. During this stage spanning from period t_0 , firm creation, to t_1 (see Figure 1), both organizational (i.e. firm's own attributes) and relational (i.e. relationship with external stakeholders) capital emerge and complement the human capital of the organization. These three types of capital will be closely linked, each of them contributing to the enhancement of the others. The entrepreneurs will have to decide the investment level (i.e. financial, time, human effort, etc.) in each of the three IC components in order to develop and strengthen firm core competencies, competencies which will render a competitive edge to the young organization.

Entrepreneurs are exposed permanently to a learning process from the moment the firm was created. They seek to reach an optimum efficiency level to become competitive in the marketplace, and often, explore new business opportunities to expand their activities. The learning, knowledge and experiences acquired in any of the three IC forms (i.e. human, organizational and relational) constitute fundamental intangible assets which will influence the survival and growth of a new organization. We should not ignore within this framework the relevance of tangible assets, because in addition to directly affecting business performance they also contribute to enriching the pool of intangibles.

The outcome of this turbulent gestation process will be the consolidation period. After few years of business activity and learning process, the entrepreneur finds out about the odds for future business success. The consolidation process of a start-up firm can take the form of a business failure (i.e. when the entrepreneurs decides to discontinue), decline (i.e. when expectations are not met and although the firm still continues, it obtains poor results), stability (i.e. when the organization has not advanced as desired, and the IC value remains as it was at inception), and growth (i.e. when the start-up firm is growing and enhancing the IC value). Any of these four outcomes will be influenced by the usefulness and value accumulated in each of the three forms of intangibles (i.e. human, organizational and relational) recognized in the IC framework.

Data and methodology

The focus of this study is the analysis of new organizations during their critical initial years. In particular, the aim of this work is to examine the influence of IC components in new firm survival and growth. For that purpose, a field study was conducted in the Basque region of Spain. Once the questionnaire was designed, data were gathered from entrepreneurs who decided to startup a business during the 1997-98 period. All the companies participating in the study have in common that they have been nurtured in a business incubator and still remain in business. From the 364 start-up projects screened by the directors from nine business incubation centers, a study sample of 114 observations was created where the unit of analysis was the start-up firm. These young companies are three to four years old and are currently struggling to overcome the gestation period.

We have classified the start-up companies of the sample into three groups: growth group (i.e. companies experiencing growth), stability group (i.e. firms which remain stable, with no growth) and decline group (i.e. companies showing declining results over time). Business performance has been measured by looking at employment, sales, and profit indicators. Following conventional IC conceptual frameworks, we include independent variables for three types of determinants of new firm survival and growth: human capital, organizational capital and relational capital (see Appendix for a more detailed description of the variables).

An exploratory analysis has been conducted in order to assess the existence of any relationship between IC variables and firm start-up performance indicators. Mean values for different capital, organizational and relational capital variables have been compared in each performance sub-sample of companies.

Results

Influence of human capital on business start-up performance

As shown in Table I, we identify three types of human capital factors: the entrepreneur's level of education, his experience and his motivation. The

	Growth $n_1 = 18$	Sales Stability $n_2 = 88$	Decline $n_3 = 8$	Growth $n_1 = 11$	Employment Stability $n_2 = 86$	Decline $n_3 = 17$	Growth $n_1 = 14$	Profit Stability $n_2 = 78$	Decline $n_3 = 22$
<i>Human capital</i>									
<i>Education</i>									
College degree (%)	66	37	25	73	41	47	57	41	50
Business courses (%)	83	49	25	82	53	35	71	53	41
<i>Experience</i>									
Entrepreneur's age (yrs)	33.4	32.2	33	36.5	32.2	30.7	31.2	32.7	32.4
Employee (yrs)	9.5	7.8	9.7	10.7	8.2	6.5	8.3	8.1	8.5
Management position (yrs)	3.4	1.5	0.2	6.3	1.4	0	1.8	1.7	1.5
Business creation (%)	11	21	12	18	22	0	7	23	9
Entrepreneur family (%)	44	45	50	45	44	53	36	50	36
<i>Motivation</i>									
Start new firm (%)	53	43	33	25	51	21	54	45	33
No job possibility (1-5)	1.8	2.1	3.6	2.8	1.9	2.8	1.8	2	2.7
Be independent (1-5)	3.9	3.8	4.4	2.9	4	3.7	3.5	3.8	4.2
No. hours/wk in new venture	53	50	42	45	50	51	56	50	46
% of own investment	56	48	53	58	48	51	61	46	57

Business start-up success

Table I.
Influence of human capital on business start-up performance

results from our sample show that the majority of companies experiencing growth in sales, employment and profit are managed by entrepreneurs with college degrees and with special interest in business education programs. Knowledge acquired in these activities seems to be an important intangible asset for building successful companies.

Prior experience in a managerial position is an important asset among companies experiencing growth. From our sample, we notice that older entrepreneurs have been employed in the past for a longer period of time, a period in which they have accumulated business experience and become acquainted with different markets. All these managerial and business experiences are important to ensure firm endurance. We expected that entrepreneurs with previous business start-up experience would perform better than those without it, since they always can avoid previous mistakes in the new venture or improve business decisions that worked out well in the past. However, results suggest that the group of firms with the largest percentage of entrepreneurs with prior start-up experience was the stability group (i.e. firms with no growth in sales, employment and profit).

Regarding the motivation component of the human capital of entrepreneurs, we observe that entrepreneurs of growing firms are on average the ones who devote a larger number of hours to their business activities (i.e. between 50 and 60 hours per week), show a more serious financial investment commitment in their ventures (i.e. about 60 percent of the venture is funded with personal savings) and are more likely to abandon the previous job in order to startup a new company. On the other hand, entrepreneurs of declining firms were characterized by having a strong need to be independent and run the business on their own. Interestingly, entrepreneurs of declining firms had serious difficulties in getting a job, and apparently, they started their firms due to the lack of any other employment alternative.

This exploratory analysis suggests that entrepreneurs' human capital, measured by knowledge, experience and motivation indicators, is an important intangible component which seems to affect new firm survival and growth. In particular, entrepreneurs with college degrees, prior managerial experience and highly committed to start a new venture were more likely to create a successful company. These results are consistent with previous findings of a large body of the literature (i.e. Bates, 1995; Hay and Ross, 1989; Cooper *et al.*, 1989; Stuart and Abetti, 1990; Doutriaux and Simyar, 1987).

Influence of organizational capital on business start-up performance

A second set of factors expected to influence new business success is organizational capital. Within this set, we considered both firms' resources and business strategies (see Table II). There are two variables related to the renewal and development component of organizational capital (Roos *et al.*, 1997), level of adaptation to the market and firm age, that appear to be crucial to explain new venture success. The variable level of adaptation to the market represents an index which captures different changes produced within a firm during the

Organizational capital	Sales			Employment			Profit		
	Growth $n_1 = 18$	Stability $n_2 = 88$	Decline $n_3 = 8$	Growth $n_1 = 11$	Stability $n_2 = 86$	Decline $n_3 = 17$	Growth $n_1 = 14$	Stability $n_2 = 78$	Decline $n_3 = 22$
<i>Firm characteristics</i>									
Adaptation (avg)	57	41	48	65	50	51	56	51	52
Market potential (1-4)	2.9	2.8	2.4	3.1	2.8	2.6	2.7	2.8	2.8
Firm age (months)	32	21	26	33	23	19	31	20	30
No. of partners	2.4	1	0.7	3.2	1.1	0.8	1.5	1	2
Initial investment ('000 US\$)	52.7	32.3	48.7	33.4	37.1	36.3	36.3	39.3	27.5
<i>Business strategy</i>									
Price strategy (1-5)	2.7	3.5	3.3	3	3.3	3.8	2.6	3.4	3.7
Quality (1-5)	4.7	4.5	4.4	4.6	4.4	4.1	4.4	4.5	4.1
Innovation (1-5)	3.9	3.5	2.8	3.5	3.6	3	3.2	3.7	3.4

Business start-up success

Table II.
Influence of
organizational capital
on business start-up
performance

gestation period in order to respond to market conditions (i.e. changes in items such as the number of new products, quality and prices of products, customers, suppliers, firm location, partners, and so on). The growing companies of the sample evince the highest adaptation index for all the performance indicators. Companies that persist in refining their strategies to adjust to new market conditions are the ones that thrive and perform best. As it is indicated by the firm age variable, mature firms have gone through a learning and transformation process for a longer period of time, which, supposedly, is expected to increase the odds for survival and growth (Birley and Westhead, 1990). Also, firms operating in industry sectors in an incipient phase of their life-cycle or in industries that show a high market potential seem to perform better than companies from other sectors. These results support previous findings by Sandberg and Hofer (1987).

In addition to organizational learning, there are other firm attributes that, although they are tangible, are important in explaining new business success. For instance, larger sums of financial resources at the moment of firm creation contribute positively to firm endurance. Furthermore, a founding team of entrepreneurs, in addition to accumulating expertise and experience within a firm, allows the expansion of sources for obtaining financial resources. Interestingly, firms that grow both in sales and employment, but obtain poor financial results, are founded by greater teams of entrepreneurs.

Regarding firm strategies, we note that companies of our sample that perform best are quality oriented, which confirms the results obtained by Kalleberg and Leicht, (1991). While owners of both stable and growing firms consider that innovation is an important strategic activity to remain in business, entrepreneurs of declining firms are the ones who value most a firm strategy based on efficiency and low pricing.

Therefore, organizational capital elements such as intra-firm learning, structural flexibility and business strategies are relevant components of the intellectual capital embedded in a firm which seem to influence new firm survival and growth. The empirical evidence about the organizational capital of the start-up firms contained in our sample indicates that firm flexibility for adapting to a market, which is perceived in most instances as a market with high potential, determines business success. This is particularly true in ventures where quality and innovation-based strategies prevail.

Influence of relational capital on business start-up performance

Relational capital is the last modality of intellectual capital analyzed in our framework (see Table III). Despite data limitations, we have explored the interaction of entrepreneurs with customers, suppliers, financial institutions, consultants, and local regulators. Due to the peculiar characteristics of our study sample, we distinguish between business networks developed by entrepreneurs out of the incubator and within the incubator.

Starting from the networks developed by the entrepreneur out of the incubator, we notice that the vast majority of successful start-up firms rely on a

Regional capital	Sales			Employment			Profit		
	Growth $n_1 = 18$	Stability $n_2 = 88$	Decline $n_3 = 8$	Growth $n_1 = 11$	Stability $n_2 = 86$	Decline $n_3 = 17$	Growth $n_1 = 14$	Stability $n_2 = 78$	Decline $n_3 = 22$
<i>Business networks</i>									
Alliances (%)	28	19	0	9	24	0	21	18	23
Small number of customers (%)	44	32	50	25	39	20	57	30	37
Large number of customers (%)	56	67	50	75	61	80	43	70	63
Small number of suppliers (%)	86	77	50	100	76	64	92	76	64
Large numbers of suppliers (%)	14	23	50	0	24	36	8	24	36
Local customers (avg)	83	71	100	81	72	87	78	72	80
Local suppliers (avg)	79	50	40	87	50	43	83	52	43
Local financial sources (avg)	91	65	75	87	72	50	87	67	73
Consultants (%)	55	50	48	54	51	46	55	51	48
Industry environment (avg)	65	63	57	64	64	58	66	63	61
<i>Local regulators, incubator</i>									
Creation (%)	72	84	75	64	81	94	79	86	68
Support (1-5)	4	4	3.7	4.1	4	3.9	4	4	4
Tangible (0-3)	1.6	1.4	1.1	1.6	1.4	1.2	1.6	1.5	1.2
Intangible (0-3)	1.4	1	0.8	1	1.1	0.6	1.4	1	0.8
Trade (0-2)	0.2	0.3	0	0	0.3	0.2	0.3	0.3	0.2
Share experiences (0-3)	1	0.8	0.4	1.1	0.8	0.7	0.9	0.8	0.8

Business start-up success

Table III.
Influence of relational capital on business start-up performance

small number of suppliers. Furthermore, these suppliers are located in the same district or province of the start-up firm. On the contrary, companies which depend mostly on local customers are showing the lowest growth and profitability performance indicators. Hence, it seems from our sample that concentrating the supplying market close to a start-up firm and expanding the sales market beyond its local area increase the odds for business success.

Certainly, owners of companies that perform best face less difficulties in obtaining additional funds from local financial institutions. Also, they consider that the advice from business consultants is helpful, and believe that the local industry environment (i.e. fiscal regulation, labor market, capital market, infrastructure, political situation) has affected favorably the evolution of their firms. These findings coincide with previous results found in the literature (Cooper *et al.*, 1989; Duchesneau and Gartner, 1990) and suggest that the ability to establish and benefit from business networks is an important intangible asset that entrepreneurs must take into account during the gestation period of a firm.

All the owners of the companies participating in this study are struggling to overcome the gestation period of their firms while they are being incubated in a business center. Our results show that the most successful entrepreneurs from our sample are the ones who value most not only the tangible services provided by the business centers (i.e. access to fax machine, photocopy and other office equipment, physical space to locate the firm), but also their intangible services (i.e. business courses, technical assistance etc.). They also value most the opportunity offered by the incubation center to share experiences and discuss business issues with other entrepreneurs hosted by the same center and living under the same roof. Obviously, the business incubation center offers a unique setting to develop an important relational capital element, such as the support climate among entrepreneurs created within the incubator to overcome together the difficult moments of the firm gestation period.

Conclusion

If research has been prominent for purposes of IC definition and classification, unfortunately, not much work has been done to better understand the influence of IC assets on business performance. IC frameworks to explain business performance may be helpful not only for large and public companies, but also for new firm start-ups. It is widely accepted that new firms play a key role in the economy and in the development of certain regions and industry sectors. Nevertheless, most of these new firms face serious problems in surviving the gestation period and perform successfully thereafter. This study has been an attempt to assess the relationship between IC intangible assets and new firm survival and growth.

Research has shown that entrepreneurs' human capital elements such as education, business experience and level of motivation are important intangible assets, which seem to be related positively to venture performance. Similarly, organizational capital elements like firm capacity to adapt quickly to changes

and the ability to implement appropriate strategies are associated with firm survival and growth. Finally, the development of productive business networks and an immediate access to critical economic agents (i.e. supply and customer markets, financial markets, local policy makers, consultants) facilitates the culmination of the gestation period. The effective interaction with all these agents and stakeholders during the initial years of business activities becomes an important relational capital element to explain new business success.

An important implication for new firm owners is that although entrepreneurs usually show special concern about the availability of tangible assets at the moment of starting up a new firm (i.e. space, financial resources etc.), they should not undervalue the significant and positive contribution of intangible assets (i.e. in the form of human, organizational and relational capital) to their companies. Similarly, policy makers should promote the enhancement of IC components in the organizations created by entrepreneurs. A business incubation center seems to be an interesting approach to nurture successful firms, a place where a specific segment of entrepreneurs receives considerable support in both tangible and intangible forms.

To conclude, further research is needed to examine the influence of IC intangible assets on business start-up performance. In this study, we have examined the relationship, not the causality, between IC elements and firm performance on a sample with limited size and characteristics. The availability of a larger sample, including firm failure observations, and the application of different statistical methods, would enhance our results.

Note

1. Motivated by the absence of a common international framework for the identification and measurement of intangibles, a team of researchers started the MERITUM project a few years ago with the goal of defining the meaning of intangibles and establishing some indications about its measurement and disclosure of information. The result of the first phase of the project has been the collection, discussion and organization of IC concepts related to the classification, measurement and report of intangibles. A no less important outcome of the project has been the dissemination of a set of guidelines for reporting intellectual capital.

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Appendix. Variables

College degree: percentage of entrepreneurs with a college degree

Business courses: percentage of entrepreneurs who attended business courses

Age: age of entrepreneurs (years)

Employee: number of years as an employee of the entrepreneur before start-up

Management position: number of years of the entrepreneur in a managerial position prior to start-up of the firm

Business creation: percentage of entrepreneurs with prior start-up experience

Entrepreneur family: percentage of entrepreneurs who belong to a family of entrepreneurs

Start new firm: percentage of entrepreneurs who abandoned the previous job to start the new venture

No job possibility: Likert scale from 1 to 5, indicating the importance level (i.e. 5 = very important) of the reason why entrepreneurs decided to create their companies

Be independent: Likert scale from 1 to 5, indicating the importance level (i.e. 5 = very important) of the reason why entrepreneurs decided to create their companies

No. hours/wk: number of hours per week devoted by the entrepreneurs to the new company

Percentage of own investment: percentage of initial capital of the new venture coming from savings of the entrepreneur

Adaptation: this variable is an index (from 1 to 100) which comprises changes (i.e. new products, markets, financial sources, prices etc.) introduced by the entrepreneurs in the company during the gestation period

Market potential: Likert scale from 1 to 4 (i.e. 4 = very high potential) indicating the growth potential of the industry sector of the start-up firm

Firm age: number of months of the company since business creation

Partners: number of founding partners at the moment of creating the firm

Initial investment: initial financial investment in thousand US\$.

Price strategy: Likert scale from 1 to 5, indicating the importance level (i.e. 5 = very important) of a low pricing strategy implemented by the entrepreneur in the organization

Quality: Likert scale from 1 to 5, indicating the importance level (i.e. 5 = very important) of a product quality strategy implemented by the entrepreneur in the organization

Innovation: Likert scale from 1 to 5, indicating the importance level (i.e. 5 = very important) of a product innovation strategy implemented by the entrepreneur in the organization

Alliances: percentage of companies with collaborative agreements with other firms

Small number of customers: percentage of firms which depended on a small number of customers

Small number of customers: percentage of firms which depended on a small size and specific customer base

Large number of customers: percentage of firms which depended on a large size and diversified customer base

Small number of suppliers: percentage of firms which depended on a small size and specific supplier base

Large number of suppliers: percentage of firms which depended on a large size and diversified supplier base

Local customers: percentage of firm sales to customers from the same province

Local suppliers: percentage of firm purchases from suppliers located in the same province

Local financial sources: percentage of firm financial capital obtained in the same province

Consultants: percentage of entrepreneurs who relied on the information provided by consultants

Industry environment: this variable is an index (from 1 to 100) which represents different aspects of the local industry environment (i.e. labour market, general economic condition, local infrastructure etc.)

Creation: percentage of entrepreneurs who stated that the business incubation center influenced their decision to create a new firm

Support: Likert scale from 1 to 5, indicating the importance level (i.e. 5 = very important) of the support received from the business incubation center

Tangible: Likert scale from 0 to 3, indicating the usefulness (i.e. 3 = very useful) of the tangible services such as physical space, access to office equipment etc. provided by the business incubation center

Intangible: Likert scale from 0 to 3, indicating the usefulness (i.e. 3 = very useful) of the intangible services provided by the business incubation center

Trade: Likert scale from 0 to 2, indicating the frequency (i.e. 2 = very often) at which entrepreneurs from the same incubator used to establish sale agreements among them

Share experiences: Likert scale from 0 to 3, indicating the frequency (i.e. 3 = very often) at which entrepreneurs from the same incubator used to share entrepreneurial experiences among them